Day 2: Audit a messy dataset

Missing values	Leakage	Label noise	Duplicates

Before you build a model, question the data. Today you will audit a loan dataset that has been deliberately damaged. Find the defects, measure what each one does to a model, and decide what to fix, drop or escalate.

Time	Reading: about 2 hours, done beforehand. Challenge: 75 to 90 minutes.
Work	Individual. Python notebook (pandas and scikit-learn) plus a written audit report.
Thinking level	Analyze. Marks go to evidence, not to opinions.
You need	loan_default_day2.csv (about 5,350 rows, 16 columns) and this handout.

The scenario

You have just joined the analytics team of a regional co-operative lender. The manager wants a model that predicts whether a borrower will default within 12 months (**defaulted_12m**). The model would be used at the moment a loan is **approved**. Before anyone trains anything, you have been asked for a data audit: is this dataset fit to build that model on, and what must change first?

The data

loan_default_day2.csv has one row per loan. It was assembled in a hurry from several systems. The descriptions below come from the lender's data team: what a column is called and what it holds may not match.

Column	Type	Description from the data team
`loan_id`	integer	Loan identifier
`applicant_name`	text	Name of the borrower
`age`	integer	Age in years
`employment_type`	text	Type of employment
`employment_years`	number	Years in current employment
`monthly_income`	number	Monthly income of the borrower, in INR
`loan_amount`	number	Amount sanctioned, in INR
`loan_date`	date	Date the loan was sanctioned
`credit_score`	integer	Bureau score, 300 to 900
`region`	text	Branch region
`collection_calls`	integer	Number of calls made by the collections team
`recovery_amount`	number	Amount recovered by the lender, in INR
`account_closed_date`	date	Date the loan account was closed, if it has been
`risk_flag`	text	Internal risk rating, high or low
`late_but_repaid`	text	Y if the borrower ever paid late but cleared dues in full, otherwise N
`defaulted_12m`	0 / 1	**Target.** 1 if the loan was recorded as defaulted within 12 months of the loan date

Step 1: Learn first

Read these before you attempt the understanding questions or the practical. They are free and online. The right-hand column shows where each one helps.

#	Resource	What to take from it	Link	Time	Helps with
1	**scikit-learn: Common pitfalls and recommended practices**	Defines data leakage (information not available at prediction time) and shows how a pipeline prevents it.	scikit-learn.org/stable/common_pitfalls.html	20 min	Q2, Q3, Part C
2	**Machine Learning Mastery: 3 subtle ways data leakage can ruin your models**	Target leakage, train-test contamination and temporal leakage, with short code showing a score inflated by a leaky feature.	machinelearningmastery.com/3-subtle-ways-data-leakage-can-ruin-your-models-and-how-to-prevent-it	20 min	Q2, Q3, Q4, Part C, D
3	**Scribbr: Missing data, types, explanation and imputation**	Explains MCAR, MAR and MNAR with examples, and why the type decides how safe it is to drop or fill values.	scribbr.com/statistics/missing-data	20 min	Q1, Part B
4	**pandas User Guide (open the topics 'Working with missing data' and 'Duplicate data')**	How pandas marks missing values, and how to fill, drop and detect them or find duplicate rows.	pandas.pydata.org/pandas-docs/version/1.5/user_guide	30 min	Parts A, B, D
5	**cleanlab documentation**	Shows how label errors, outliers and near-duplicates can be found automatically. Read the idea; you may do it by hand or with the tool.	docs.cleanlab.ai/v2.6.5	25 min	Q5, Part D

If a link does not open, search for the resource title. Pages on other sites may change over time.

Step 2: Understanding questions

Answer these on Answer Sheet A, in 3 to 5 lines each and in your own words. Do them **before** opening the dataset. They are worth 5 marks.

No.	Question	Read first
Q1	What is the difference between data that is missing at random and data that is missing not at random? Give an example from a lending setting, and say which one is more dangerous to ignore.	Resource 3
Q2	Why is a very high validation score sometimes a warning sign and not good news?	Resources 1, 2
Q3	What single question separates a legitimate feature from a leaky one? Apply it to two features you invent for a loan-approval model.	Resources 1, 2
Q4	Why can duplicate rows make a model look better than it really is? Where in the workflow must duplicates be handled?	Resource 2
Q5	Why should label noise be investigated before tuning hyperparameters?	Resource 5
Q6	When is it right to drop a row, and when is it right to ask the data owner instead?	Resources 3, 4

Step 3: The audit

Work through Parts A to E in order. Keep a notebook, and write down every number you rely on. Do not edit the CSV file. Make every change in code so that someone else can repeat it. Convert text columns to numbers before modelling, fill blanks in a way that cannot leak, and for each finding write what you did, what you saw, and what you decided.

Part A: First look (about 10 min)

- Check shape, data types, and the range of every numeric column. Run `describe()` and count the values in each text column.
- Write down five things that look odd **before you change anything**.

Think: Which odd things are errors, and which are just unusual but real?

Part B: Missing values (about 15 min)

- Find the percentage missing in every column. Include blanks that are hiding behind placeholder values.
- Compare the default rate for rows with and without a monthly income.
- Choose and justify a treatment for monthly_income.

Think: What would filling every blank with the average do to the story the data is telling?

Part C: Leakage hunt (about 20 min)

- Go through every feature and ask whether you would know it at the moment the loan is approved.
- Train one baseline model using all features and report its 5-fold cross-validated AUC.
- Remove suspicious features one at a time and record how the score changes.

Think: If the first model had scored only 0.70, would you still have found the leaks? What is the warning sign?

Part D: Duplicates and label noise (about 20 min)

- Count exact duplicates. Then find near-duplicates: the same person recorded twice with small differences in formatting.
- Show the effect of duplicates: compare a score with them left in against a score after removing them.
- Find rows where the label disagrees strongly with the other evidence, and inspect 20 of them by hand.
- Check whether the labelling rule for late-but-repaid loans is applied consistently.

Think: Could a model trained on this data learn the labelling habits of a region and not the behaviour of borrowers?

Part E: Audit report (about 15 to 20 min)

- List every issue you found. Give each one a severity (high, medium or low) and an action: fix, drop, ask the data owner, or monitor.
- Finish with a one-paragraph recommendation: is this data ready for modelling, yes or no, and what must happen first?

Think: Which two issues would you fix first if you had only one day?

Starter code (adapt it, do not copy it blindly)

```
import pandas as pd, numpy as np
from sklearn.model_selection import cross_val_score
from sklearn.ensemble import GradientBoostingClassifier

df = pd.read_csv('loan_default_day2.csv')
df.isna().mean() # blanks per column
df.duplicated().sum() # exact duplicate rows
df.groupby(df['monthly_income'].isna())['defaulted_12m'].mean()

# cross-validated AUC for a chosen list of columns
cross_val_score(GradientBoostingClassifier(), X, y, cv=5, scoring='roc_auc')
```

Step 4: What to submit

Answer Sheet A	Your answers to the six understanding questions
Answer Sheet B	Table 1 (missing values by column) and Table 2 (leakage log)
Answer Sheet C	Table 3 (duplicates) and Table 4 (twenty inspected labels)
Answer Sheet D	Your audit report: issues ranked by severity, with an action for each, and your final recommendation
Notebook	Runs from top to bottom without errors and produces every number you reported

How you will be marked (15 marks)

Criterion	Marks
Understanding questions Q1 to Q6	5
Finds hidden missing values and tests whether missingness is informative	2
Identifies leakage using the timing test, with before and after evidence	3
Detects exact and near duplicates and shows the effect on the score	2
Investigates label noise by inspecting real rows, not guessing	1
Audit report is prioritised and every action is justified	2
Total	**15**

Evidence wins. A claim such as 'income is leaky' scores less than one that shows the score before and after removal. A suspicion you could not prove is still worth reporting, if you say it is only a suspicion.

Rules

- Work on your own. You may talk about ideas, but your notebook and reasoning must be your own.
- Do not edit the original CSV. Every change must be written as code and listed in your report.
- You may be asked to explain any finding aloud, and to re-run your notebook.
- Tools that profile data automatically are allowed, but you must say what they found and what they missed.
- Late submissions are handled as announced by your instructor.

Submission

Submit by: _____

Submit to: _____

Name: _____ Roll number: _____ Batch: _____

Answer Sheet A: Understanding questions

Name: _____ Roll no.: _____

Q1. What is the difference between data that is missing at random and data that is missing not at random? Give an example from a lending setting, and say which one is more dangerous to ignore.

Q2. Why is a very high validation score sometimes a warning sign and not good news?

Q3. What single question separates a legitimate feature from a leaky one? Apply it to two features you invent for a loan-approval model.

Q4. Why can duplicate rows make a model look better than it really is? Where in the workflow must duplicates be handled?

Q5. Why should label noise be investigated before tuning hyperparameters?

Q6. When is it right to drop a row, and when is it right to ask the data owner instead?

Answer Sheet B: Missing values and leakage

Name: _____ Roll no.: _____

Table 1: Missing values by column

Column	% missing (include placeholders)	How you found it	Random or not? (MCAR, MAR, MNAR)	Your treatment

Table 2: Leakage log

Feature	Why it fails the timing test	AUC before removal	AUC after removal	Decision

Answer Sheet C: Duplicates and label noise

Name: _____ Roll no.: _____

Table 3: Duplicates

Item	Your result
Exact duplicate rows found	
Near-duplicate rows found	
How you detected near-duplicates	
Score with duplicates left in (state the metric)	
Score after removing duplicates	

Table 4: Twenty rows you inspected by hand

#	Row id (loan_id)	Given label	Why it looks suspicious	Your verdict (noise, real, unsure)
1				
2				
3				
4				
5				
6				
7				
8				
9				
10				
11				
12				
13				
14				
15				
16				
17				
18				
19				
20				

Answer Sheet D: Audit report

Name: _____ Roll no.: _____

Rank	Issue	Family	Evidence (numbers)	Severity H/M/L	Action	Reason
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						

Final recommendation: is this data ready for modelling? (Yes or no, and what must happen first.)